

AI-Powered Collaborative Task Management and Team Productivity Platform

Problem Statement

Context:

Modern teams require efficient platforms to manage projects, collaborate on tasks, monitor progress, and make data-driven decisions. Traditional task management systems mainly provide basic task creation, assignment, and tracking, but they lack intelligent capabilities to automatically prioritize work, predict delays, recommend task assignments, and assist team members in their daily activities.

Objective:

Develop a **full-stack AI-powered collaborative task management platform** using the **MERN stack** that enables users to create projects, assign and manage tasks, collaborate in real-time, monitor progress, and use AI-based features for intelligent task prioritization, recommendations, notifications, and productivity analysis.

Key Features

1. User Authentication & Team Management

- User registration, login, logout, and profile management.
- Role-based access for administrators, project managers, and team members.
- Create and manage teams and projects.

2. Intelligent Task Management

- Create, assign, update, and track tasks.
- Set priorities, deadlines, dependencies, and task status.
- AI-based **task priority and urgency recommendations**.
- AI-generated task descriptions and subtasks from simple user requirements.
- AI recommendations for suitable team members based on skills and workload.

3. Real-Time Collaboration

- Real-time task updates using **WebSockets/Socket.IO**.
- Team chat and task-specific discussions.
- Document/file sharing.
- Real-time activity feed.

4. AI Productivity Assistant ☐

- AI chatbot/assistant for project and task-related queries.
- Convert natural-language instructions into tasks.
- Summarize project discussions and task updates.
- Generate meeting/action-item summaries.

- Recommend the next tasks to be completed.
- Provide intelligent suggestions for improving team productivity.

5. Progress Tracking & Predictive Analytics

- Real-time task and project progress tracking.
- Interactive dashboards, progress bars, Kanban boards, and Gantt charts.
- AI-based prediction of potentially delayed tasks.
- Identify overdue and high-risk tasks.
- Generate project performance insights.

6. Smart Notifications

- Notifications for task assignments, updates, deadlines, and dependencies.
- Email and in-app notifications.
- AI-assisted notification prioritization to highlight important actions.

Key Challenges

1. **Full-Stack Integration:** Integrating React, Node.js, Express, MongoDB, REST APIs, and WebSockets into a scalable application.
2. **AI Integration:** Integrating an AI/LLM API to provide intelligent recommendations, summarization, natural-language task creation, and conversational assistance.
3. **Real-Time Data Synchronization:** Maintaining consistent task, chat, and notification data across multiple users.
4. **Scalability & Security:** Supporting multiple teams and projects while implementing authentication, authorization, secure APIs, and efficient database operations.
5. **User Experience:** Designing a responsive, intuitive dashboard for web and mobile devices.

Learning Objectives

By completing this project, students will:

1. Gain practical experience in **MERN-based full-stack application development**.
2. Design and implement **RESTful APIs and database-driven applications**.
3. Implement **authentication, authorization, and secure application development**.
4. Integrate **WebSockets/Socket.IO for real-time communication**.
5. Integrate **AI/LLM APIs into a full-stack application**.
6. Implement AI-based features such as **recommendation, summarization, prediction, and conversational assistance**.
7. Develop interactive dashboards and data visualizations.
8. Understand deployment, scalability, testing, and documentation of full-stack applications.

Deliverables

1. **Fully functional AI-Powered Collaborative Task Management Platform** deployed on a cloud platform.
2. Complete **GitHub source-code repository** with proper documentation.
3. Database design, API documentation, and system architecture.
4. **User Manual** explaining system functionality.
5. **Demo Video** demonstrating both full-stack and AI features.
6. Technical report covering:
 - Problem definition
 - System architecture
 - Technology stack
 - Database design
 - API implementation
 - AI integration
 - Testing
 - Challenges and solutions
 - Results and future enhancements

□ Alternative Titles

If this is being given as a **major Full Stack Development & AI project**, I particularly recommend one of these:

1. **AI-Powered Collaborative Task Management and Team Productivity Platform** □
2. **IntelliTask: An AI-Driven Collaborative Project Management System**
3. **SmartWork: AI-Powered Team Collaboration and Task Management Platform**
4. **AI-TaskHub: Intelligent Project and Team Management System**
5. **Collaborative AI-Powered Project Management and Productivity Platform**

My recommendation: IntelliTask – AI-Powered Collaborative Task Management and Team Productivity Platform. It sounds more like a modern **Full Stack + AI** project while still allowing students to demonstrate MERN, APIs, WebSockets, databases, authentication, cloud deployment, and practical AI integration.